Human Computer Interaction
Course COMP322

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Section:1

2. Usability Test Protocol

2.1 Objective:

The purpose of this usability test is to evaluate the redesigned QOU Student Portal prototype and determine whether the new design improves usability, reduces cognitive load, and enables students to complete common academic tasks efficiently. The evaluation focuses on the main redesign goals identified in Phase 1, including easier access to schedules, grades, messages, and course services.

2.2 Participants:

Three university students who did not participate in the Phase 1 interviews were recruited for the usability test. All participants had previous experience using university portals and represented different academic backgrounds.

Participant	**Age**	**Major**	**Experience with University Portals**
P1	20	Computer Science	Frequent User
P2	21	Business Administration	Moderate User
P3	22	Engineering	Frequent User

2.3 Testing Environment

- Device: Laptop/Desktop
- Prototype: Interactive Figma Prototype of the redesigned QOU Student Portal
- Method: Think-Aloud Protocol
- Duration: Approximately 15 minutes per participant

Participants were asked to verbalize their thoughts, expectations, and difficulties while completing the assigned tasks.

2.4 Testing Procedure

1. Participants were introduced to the purpose of the study and signed the consent form.
2. Each participant used the interactive prototype individually.
3. Participants completed a set of predefined tasks while following the think-aloud protocol.
4. The evaluators recorded task completion, completion time, errors, and participants' comments during the session.
5. At the end of the session, participants completed the System Usability Scale (SUS) questionnaire.
6. Quantitative data (task completion, time on task, and error counts) and qualitative observations (participants' comments and difficulties) were collected and analyzed.

2.5 Test Tasks

Participants were asked to complete the following tasks using the redesigned QOU Student Portal prototype while thinking aloud during the session.

Task 1: Check Today's Information

Scenario: You have just logged into the portal and want to know what is important today.

Task: Find today's classes and check whether there is any new message or grade update.

Success Criteria: The participant successfully identifies today's classes and notices any new updates.

Task 2: View Next Week's Schedule

Scenario: You want to know your classes for next week.

Task: Open the Semester Schedule page and display the schedule for next week.

Success Criteria: The participant successfully changes the schedule view to next week.

Task 3: Find Course Details

Scenario: You need additional information about a course.

Task: Open the Semester Schedule page and view the details of the course "Electrical Circuits".

Success Criteria: The participant successfully opens the course details pop-up and identifies the course information.

Task 4: Check Course Grades

Scenario: You want to check your performance in a course.

Task: Open Course Services, select "Electrical Circuits", and find your Overall Grade and Class Average.

Success Criteria: The participant successfully identifies both the overall grade and class average.

Task 5: Check Assignment Status

Scenario: You want to know whether you still have pending assignments.

Task: Open the Assignments tab and identify which assignments are marked as "Not Submitted".

Success Criteria: The participant correctly identifies all assignments that have not been submitted.

Task 6: Access Learning Materials

Scenario: You need to access the learning materials for one of your courses.

Task: Open Course Services, navigate to the Links tab, and access the Course Material link.

Success Criteria: The participant successfully reaches the Course Material link.

Task 7: Check Upcoming Exams

Scenario: You want to know when and where your next exam will take place.

Task: Open Course Services, select "Electrical Circuits", and find the date, time, and location of the next exam.

Success Criteria: The participant successfully identifies the next exam and correctly reports its date, time, and location.

2.6 Quantitative Results

To evaluate the usability of the redesigned QOU Student Portal, quantitative measures including task completion rate, time on task, and error count were collected for each participant.

Time on Task (seconds)

Task	P1	P2	P3	Average
Task 1	18	20	17	18.3
Task 2	24	26	22	24.0
Task 3	21	24	20	21.7
Task 4	32	35	29	32.0
Task 5	27	30	26	27.7
Task 6	19	22	18	19.7
Task 7	23	25	21	23.0

Task Completion Rate

Task	Completion Rate
Task 1	100%
Task 2	100%
Task 3	100%
Task 4	100%
Task 5	100%
Task 6	100%
Task 7	100%

Error Count

Task	Total Errors
Task 1	0

Task	**Total Errors**
Task 2	1
Task 3	0
Task 4	2
Task 5	1
Task 6	0
Task 7	1

The results indicate that all participants were able to successfully complete all tasks using the redesigned prototype, resulting in a task completion rate of 100%. The average task completion time ranged from 18.3 to 32.0 seconds, with the grades task requiring the longest time due to the additional information presented. The number of errors was low across all tasks, suggesting that the redesigned interface was easy to understand and navigate.

2.7 Qualitative Findings

Several themes emerged from the usability test sessions.

Theme 1: Easier Access to Information

All participants appreciated having important academic information available directly on the dashboard. They were able to identify today's classes and recent updates quickly.

P1: "Everything I need is visible immediately."

P3: "I do not need to search through different menus anymore."

Theme 2: Improved Schedule Navigation

Participants found the redesigned schedule easier to understand because the portal automatically displayed the current week and allowed them to switch between weeks without calculating odd and even weeks.

P2: "This is much better than calculating odd and even weeks."

Theme 3: Clear Course Services Organization

Participants were able to move easily between Exams, Assignments, Grades, and Links within Course Services. The tab structure reduced navigation effort and made information easier to find.

P1: "The tabs are easy to understand and everything is organized."

Theme 4: Better Understanding of Grades and Assessments

Participants appreciated seeing grades and class averages presented clearly. They reported that the redesigned interface helped them understand their academic performance more quickly.

P3: "I can understand my performance immediately without opening multiple pages."

Overall, participants reported that the redesigned portal was simple to use, required fewer navigation steps, and reduced the effort needed to access important academic information.

2.8 System Usability Scale (SUS) Results

At the end of the usability testing session, participants completed the System Usability Scale (SUS) questionnaire to evaluate their overall experience with the redesigned QOU Student Portal.

Participant	SUS Score
P1	85
P2	82.5
P3	87.5

Average SUS Score:

[
\frac{85 + 82.5 + 87.5}{3} = 85
]

The redesigned QOU Student Portal achieved an average SUS score of **85 out of 100**. According to the SUS interpretation scale, a score above 80 indicates **excellent usability** and high user satisfaction. The results suggest that participants found the redesigned portal easy to learn, simple to navigate, and efficient for completing common academic tasks. The high SUS score also indicates that the redesign successfully reduced cognitive load and improved access to schedules, grades, messages, and course services.

3. Heuristic Re-evaluation

A heuristic re-evaluation was conducted to determine whether the redesigned QOU Student Portal successfully addressed the usability problems identified during Phase 1. The redesigned prototype was evaluated using Nielsen's Ten Usability Heuristics and showed significant improvements in usability, navigation efficiency, and information accessibility.

Phase 1 Problem	Violated Heuristic(s)	Redesign Improvement	Result
Dense schedule tables containing unnecessary information and irrelevant columns	H8: Aesthetic and Minimalist Design	Irrelevant information was removed and additional details were moved into course details pop-ups.	Resolved
Students had to remember whether the current week was odd or even	H2, H6, H7, H8	The system automatically displays the current week's schedule and provides a week selector for future schedules.	Resolved
Course services were repeated for every course, creating	H8	A single Course Services section with a course selection dropdown	Resolved

Phase 1 Problem	Violated Heuristic(s)	Redesign Improvement	Result
long and crowded pages		was introduced.	
Important information such as meeting status and online sessions was unclear	H1, H10	Clear status messages and explanations were added for unscheduled or online meetings.	Improved
Students had to open every exam and assignment panel to check for updates	H1, H5, H7	Status indicators and notifications were added to highlight new exams and assignments.	Resolved
Exam schedules required filtering before displaying information	H7, H8	Upcoming exams are displayed immediately and filtering became optional.	Resolved
Exam review functionality produced error messages without	H5, H7	Exam review availability is now clearly indicated using enabled and disabled states.	Resolved

Phase 1 Problem	Violated Heuristic(s)	Redesign Improvement	Result
status indication			
Students had to open every course individually to check whether grades had been published	H1, H5, H7	Grade publication status is now visible through dashboard indicators and status labels.	Resolved
Assessment names were static and difficult to understand	H2, H10	Assessment names became dynamic and specific to each course.	Improved

The heuristic re-evaluation demonstrates that the redesigned portal successfully addressed the major usability problems discovered during Phase 1. The redesign reduced cognitive load, improved visibility of system status, prevented user errors, and provided a more efficient and intuitive user experience.

4. Revision Plan

Although the usability evaluation showed that the redesigned QOU Student Portal achieved high usability and positive user feedback, several improvements could still be implemented in future iterations.

First, dashboard customization could be introduced to allow students to prioritize the information that is most relevant to them, such as grades, assignments, or upcoming exams.

Second, push notifications could be added to provide immediate alerts when grades are published, new assignments become available, or exam schedules change.

Third, additional filtering and search options could be provided in the Schedule and Course Services sections to help students locate information more quickly, especially when managing multiple courses.

Finally, contextual help and tooltips could be expanded to explain academic terminology and system features for new students who may not be familiar with the portal.

These improvements would further reduce cognitive load, improve efficiency, and provide a more personalized and accessible user experience.

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